

How Open Source Software Can Help in Disaster Management

Climate change is gathering momentum and because of this, the world has witnessed some of the most devastating disasters recently. Tsunamis, cyclones, floods, earthquakes and even wildfires have destroyed lives and entire regions, of late. So can we look to open source software to help mitigate the effects of these disasters?



Web development frameworks

One of the first areas in which computing technology was used during the floods was in Web development. Government agencies, NGOs, educational institutions, etc, either made changes to their existing websites or created new ones to tackle the problems faced during the floods. There were a lot of requests for the services of Django developers on various social media platforms during the floods—far more than for any other Web development framework. Django is a free and open source Web framework, developed using Python. The reason for Django's popularity during the flood is that it supports a rapid development model and thus is ideal for getting websites up and running quickly.

In moments of crisis, people may not care much about the look and feel of the websites, but rather about the services provided. Thus, in such situations, Django is the Web development framework to depend on. But post the disaster, environment tools like AngularJS and Node.js can also be used — both are open source Web development frameworks based on JavaScript. Node.js is mainly used to build server-side applications, whereas AngularJS is suited for building single-page client-side Web applications. I believe if different disaster management agencies have a pool of volunteer Web developers available to be deployed immediately, then the crises due to natural calamities can be managed more efficiently.

Last August, Kerala reeled under one of the most severe floods it has witnessed in over a century. In one way or another, the entire population of the state (about 33 million) was affected by this disaster. More than a million people were evacuated from their dwellings and about 500 lost their lives. The amount of damage to property and livestock is yet to be fully assessed. Though a tragedy of epic proportions, it clearly brought out the best in people. The selfless deeds of countless individuals are still widely circulated on social media. The 2018 Kerala floods also saw the widespread use of technology for disaster management.

Rescue personnel on the ground often lack information regarding the locality and the people affected during natural calamities. Communication and computation technology comes handy in such situations to provide the necessary data required for rescue and evacuation. Social media platforms

like Facebook, WhatsApp, Twitter, etc, were widely used to locate and track people stranded in isolated locations due to the flood waters. Many people who were often criticised for their excessive use of social media platforms became heroes overnight because of their actions that saved countless lives. Those who were stranded in isolated locations, requested help through social media platforms and these distress calls were then verified by volunteers. Once the authenticity of a request was confirmed, it was reported to the personnel responsible for relief work. This sort of timely intervention by a large volunteer force mitigated the severity of the floods to a great extent even with large scale power outage and other logistical nightmares. In this article we will discuss ways in which computing technology was used for disaster management during the floods and also some of the other open source tools ideal for disaster management.